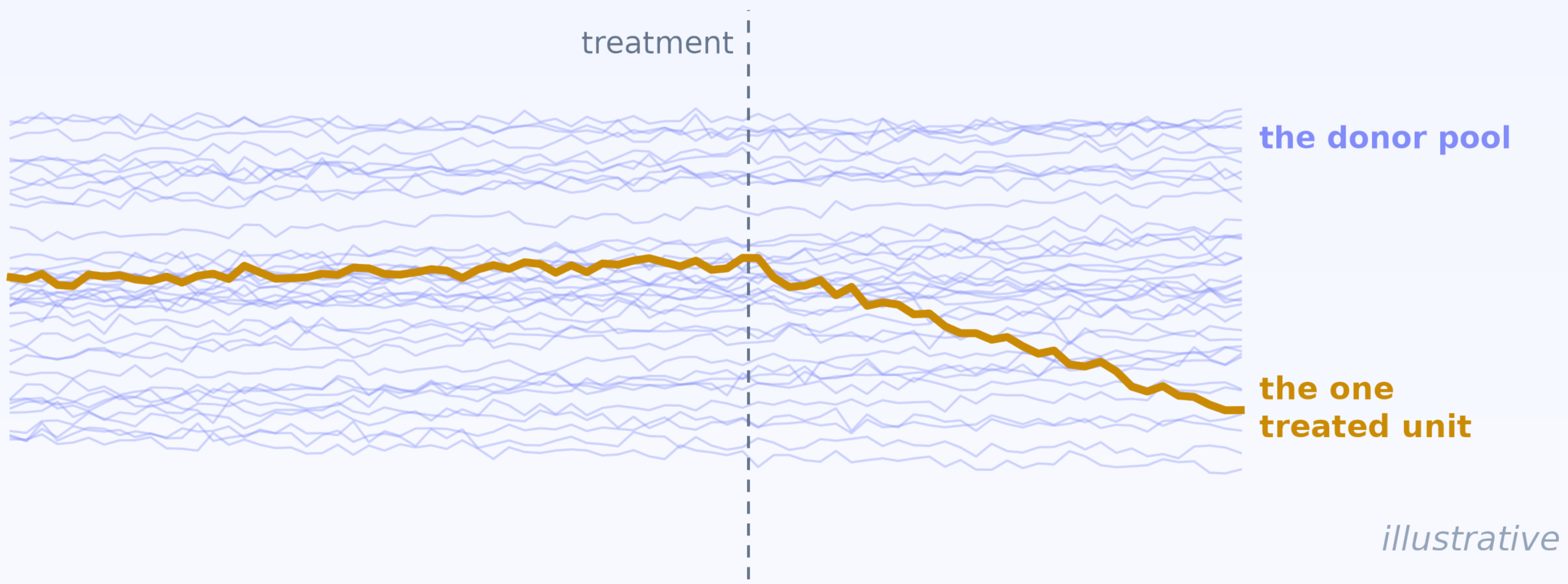
diff - diff

One treated unit.

An exact p-value.



New in diff-diff 3.10: the LWDiD estimator.

Rolling-transformation DiD (Lee & Wooldridge).

WHY ANOTHER ESTIMATOR

An estimate with one treated unit is easy.

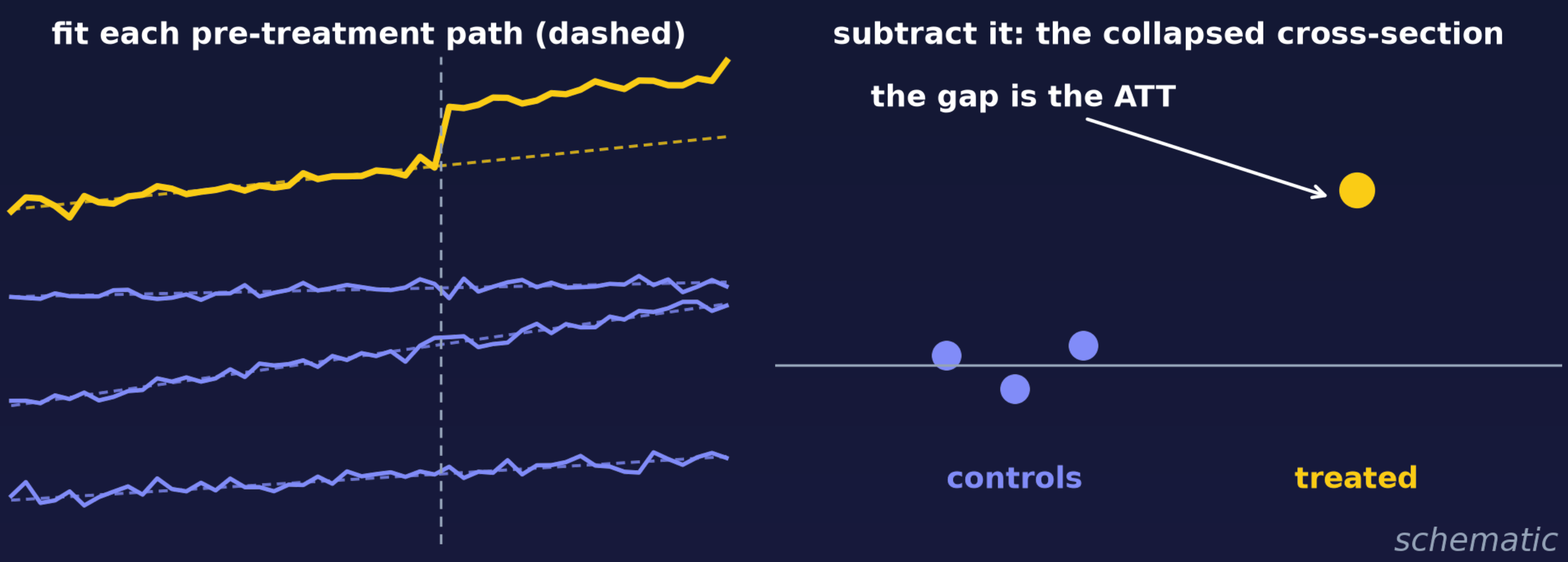
SyntheticDiD does it.

An exact p-value with one treated unit is not.

LWDiD closes that gap on its classical path: one ordinary cross-sectional regression.

THE TRICK

Subtract each unit's own pre-treatment path.



What's left is a cross-section - where exact inference works with one treated unit.

Assumes the DiD design assumptions (no anticipation, the parallel- or unit-linear-trends restriction, overlap) plus classical errors: independent, mean-zero, conditionally normal, homoskedastic in the collapsed cross-section. One treated unit needs at least two controls (N >= 3).

PROOF WITH ONE TREATED UNIT

Proposition 99:

California vs. 38 states.

p = 0.021

exact t

df = 37

p = 0.054

randomization

inference

ATT -0.227 (SE 0.094) on log cigarette sales -

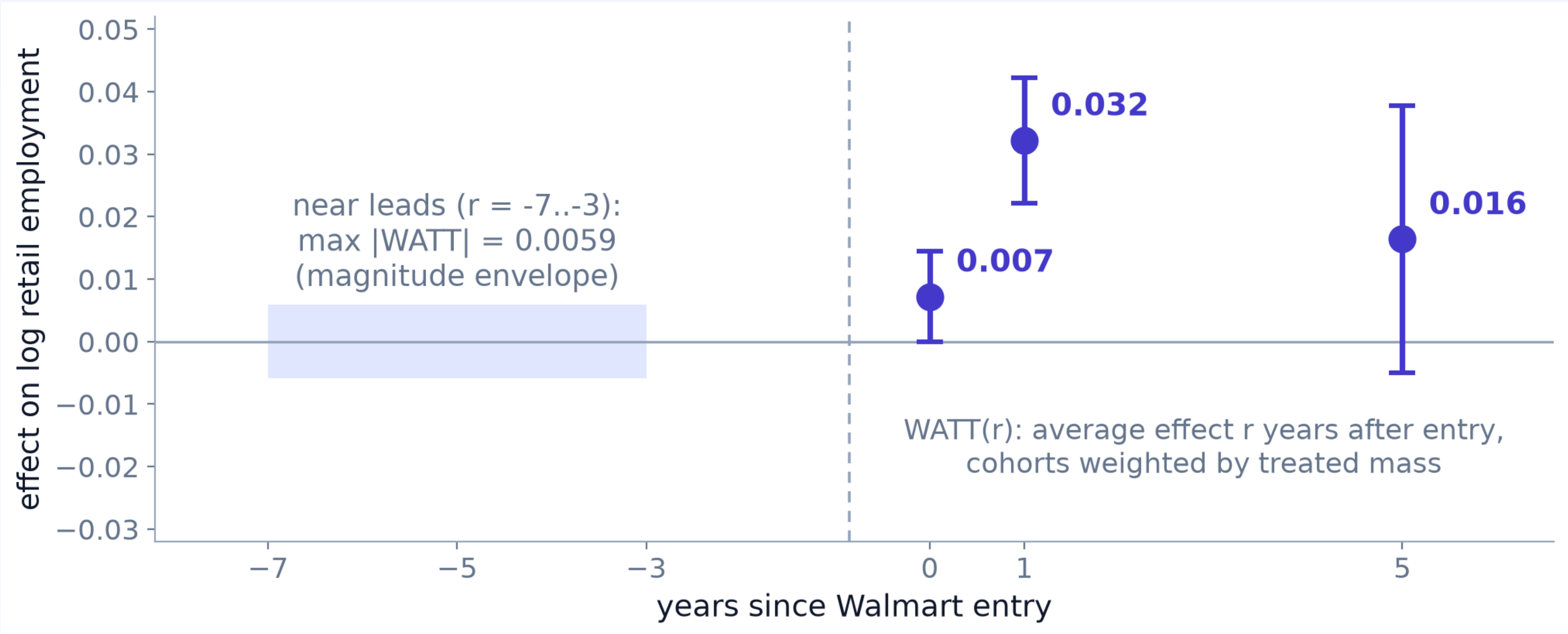
about a 20% reduction.

39 states x 31 years (1970-2000), detrended; ATT, SE, and exact p reproduce LW 2026 Table 3. RI tests Fisher's sharp null under the complete-randomization assignment mechanism (authors' package convention). Two roads to inference with a single treated state - conditional on the detrending (CHT) and design assumptions; the exact t additionally assumes independent, conditionally normal, homoskedastic collapsed errors.

AND AT SCALE

1,277 counties.

14 Walmart entry cohorts.



overall ATT 0.0199 (SE 0.0090)

Staggered adoption (1986-1999), 391 never-treated counties, detrended. Near leads stay below |WATT| = 0.006; distant leads reach |WATT| = 0.033 (at r = -22).

Simulation check: staggered ATT 2.953 (SE 0.060) vs truth 2.87 (simulated).

USE IT WELL

Units drifting apart? Detrend.

Detrending fits unit-specific linear drift. Under drift, demeaning estimated 3.38 vs detrending 1.10 - truth 1.0 (simulated). On Prop 99 the two disagree by nearly a factor of two (-0.42 vs -0.23).

Shared shocks? Cluster.

The naive SE was barely half the honest one: 0.131 unclustered vs 0.237 CR1 with G = 12 regions (simulated). Few clusters? Wild cluster bootstrap is built in for clustered common-timing reg fits: p = 0.0025, 95% CI [0.503, 1.614].

Inspect the transformation - descriptive, not an assumption test:

```
get_transformation_diagnostics()
```

THE WHOLE WORKFLOW

A fit, an exact p-value, a permutation test.

```
fit = LWDiD(rolling="detrend",
            vcov_type="classical").fit(
    prop99, outcome="lcigsale",
    unit="state", time="year",
    treatment="treated")

fit.p_value        # exact t, df = 37
fit.randomization_test(n_reps=9999, seed=42)
```

Staggered? Same fit - just add first_treat=.

The randomization test is common-timing reg only; most staggered aggregates use influence-function or bootstrap inference (an eligible classical composite keeps exact t).

Panels in.

Cross-sections out.

```
pip install diff-diff
```

Tutorial 31 reproduces every number on this deck - Prop 99 and Walmart included:

diff-diff.readthedocs.io/en/latest/tutorials/31_lwdid.html

github.com/igerber/diff-diff

Lee & Wooldridge (2025; 2026) - SSRN 4516518 + 5325686.